

William O'Connell

Quantitative Trading Analyst & AI Engineer / Rates · Credit · Equities

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EDUCATION

University of Waterloo

Statistics, Honours, Co-operative Program — concurrent Applied Mathematics (Scientific Computing & Scientific Machine Learning), Honours

Co-op · five work terms across capital markets desks

University of Chicago

Summer Program — Economics

Jul 2022 · Applied Economic Analysis · Experimental Economics Award for best performance in applied economic experimentation

EXPERIENCE

RBC Capital Markets

Sep 2026 – Dec 2026 (incoming)

Quantitative Trading Analyst & AI Engineer — Global Edge Program · New York, NY

RBC Capital Markets

Jan 2026 – Apr 2026

Quantitative Trading Analyst & AI Engineer — Municipal Systematic, Lev Fin, EQD Flow, IG Credit, Rates, Money Markets

- Built a sales recommendation engine predicting client demand for credit trade axes from historical flow and interaction data, flagging relative value to factor indices to surface high-conviction trade ideas
- Built a commission reconciliation platform that automated real-time error detection and counterparty dispute notifications — recovered \$1.5M/year in previously written-off commissions
- Built a bond cost basis interpolator estimating holding costs across SMA and public account portfolios for the Municipal Systematic Trading desk

BMO Capital Markets

Sep 2025 – Dec 2025

Global Markets Quantitative Analyst — Rates Sales, Strategy & Trading

- Built a bond duration & roll-date framework generating cross-market CAD trade ideas from historical trend analysis
- Built markout models in Python over historical trade data to detect adverse flow and support desk-level risk decisions
- Built an integrated trading dashboard (Power BI, DAX + VBA) combining multiple SQL sources to track rolls across bond forwards, repos, and TRS

CIBC Capital Markets

Jan 2025 – Apr 2025

GM & IB Analyst — Securitization & Financial Institutions Group

- Built a real-time conduit tracking tool (VBA) monitoring commitments and funding allocations across a \$10B+ deal portfolio

CIBC Capital Markets

May 2024 – Aug 2024

Global Markets Analyst — eFICC Trading

- Built a PnL tracking suite (VBA) consolidating fiscal YTD algo and WM/Reuters PnL, plus 20+ SQL/Tableau views on PnL and trade delta

PROJECTS

swaption-pricer — github.com/WilliamO-C/swaption-pricer

From-scratch USD swaption pricing library: a real bootstrapped U.S. Treasury curve, Black-76, the Bachelier normal model, and full SABR smile calibration, with closed-form Greeks. 36/36 tests passing.

SKILLS

Languages	Python, SQL, R, VBA, DAX
Libraries	NumPy, Pandas, SciPy, Matplotlib
Platforms	Power BI, Tableau, Bloomberg Terminal, Git, Jupyter
Data	S&P Capital IQ, Fitch, Moody's, DBRS